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0.1 Question 1.6

Show that

$$\sin(\frac{2\pi}{10}) + \sin(\frac{4\pi}{10}) + \cdots + \sin(\frac{10\pi}{10}) = \sqrt{\frac{1}{2}(5-\sqrt{5})} + \sqrt{\frac{1}{2}(5+\sqrt{5})}$$

Lets say LHS $= \sin(\frac{2\pi}{10}) + \sin(\frac{4\pi}{10}) + \cdots + \sin(\frac{10\pi}{10})$ and RHS $= \sqrt{\frac{1}{2}(5-\sqrt{5})} + \sqrt{\frac{1}{2}(5+\sqrt{5})}$

Using the longer method…

[1]:
```python
import math
import numpy
import sympy

sympy.sin(2 * sympy.pi/10) + sympy.sin(4 * sympy.pi/10) + sympy.sin(6 * sympy.
 ↪pi/10) + sympy.sin(8 * sympy.pi/10) + sympy.sin(10 * sympy.pi/10)
```

[1]:

$$2\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}} + 2\sqrt{\frac{\sqrt{5}}{8} + \frac{5}{8}}$$

From here we can see that using the `sympy` library we have simplified the LHS equation to $2\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}} + 2\sqrt{\frac{\sqrt{5}}{8} + \frac{5}{8}}$, lets see if we use either numpy or math library.

[2]:
```python
num = numpy.sin(2 * numpy.pi/10) + numpy.sin(4 * numpy.pi/10) + numpy.sin(6 *␣
 ↪numpy.pi/10) + numpy.sin(8 * numpy.pi/10) + numpy.sin(10 * numpy.pi/10)
print(f"The numpy value of num is {num}")
maths = math.sin(2 * math.pi/10) + math.sin(4 * math.pi/10) + math.sin(6 * math.
 ↪pi/10) + math.sin(8 * math.pi/10) + math.sin(10 * math.pi/10)
print(f"The math value of maths is {maths}")
```

```
The numpy value of num is 3.0776835371752536
The math value of maths is 3.0776835371752536
```

From the 2 inputs above which have the `numpy` and `math` libraries, both have given the same answer of 3.07768… So assuming that RHS is would give the same answer as the one in LHS.

[3]:
```python
a = (5 - math.sqrt(5))
print(f"The value of 5-math.sqrt is {a}")
b = (5 + math.sqrt(5))
print(f"The value of 5+math.sqrt is {b}")
```

```
The value of 5-math.sqrt is 2.76393202250021
The value of 5+math.sqrt is 7.23606797749979
```

[4]:
```
math.sqrt(1/2 * a) + math.sqrt(1/2 * b)
```

[4]:
```
3.077683537175253
```

There we can see that both LHS = RHS as both are 3.07768... using the `math` library. Lets find out the one using the `sympy` library.

[5]:
```
a = (5 - sympy.sqrt(5))
a
```

[5]: $5-\sqrt{5}$

[6]:
```
b = (5 + sympy.sqrt(5))
b
```

[6]: $\sqrt{5}+5$

[7]:
```
sympy.sqrt(1/2 * a) + sympy.sqrt(1/2 * b)
```

[7]: $\sqrt{2.5-0.5\sqrt{5}}+\sqrt{0.5\sqrt{5}+2.5}$

Although the equation would look different, they still give the same answer, so ∴ LHS = RHS.

Now using the shorter method

[8]:
```
left = numpy.sin(2 * numpy.pi / 10) + numpy.sin(4 * numpy.pi / 10) + numpy.
↪sin(6 * numpy.pi / 10) + numpy.sin(8 * numpy.pi / 10) + numpy.sin(10*numpy.
↪pi/10)

a = (5 - numpy.sqrt(5))
b = (5 + numpy.sqrt(5))
right = (numpy.sqrt(1/2 * a) + numpy.sqrt(1/2 * b))

isequal = numpy.isclose(left, right)

print(f"Is the LHS = {left} equal to RHS = {right}?")
print(f"{isequal}, LHS = RHS")
```

```
Is the LHS = 3.0776835371752536 equal to RHS = 3.077683537175253?
True, LHS = RHS
```

To find out that LHS = RHS, I tried out a shorter method where I just had to use the `numpy.isclose` function to check if both the `left` and `right` are equal.

0.2 Question 4.10

Using functional programming, demonstrate that for any sequence of positive numbers $a_1, a_2, \dots, a_n$ we have

$$(a_1+a_2+\cdots+a_n)(\frac{1}{a_1}+\frac{1}{a_2}+\cdots+\frac{1}{a_n})\geq n^2$$

Here we use the `lambda` function or the anonymous function to figure out the sequence.

[9]:
```
from functools import reduce

number = int(input("Enter a number "))

def product(x, y):
    return x * y
number_digits = [int(digit) for digit in str(number)]

a = reduce(lambda x, y: x + y, number_digits)
b = reduce(lambda x, y: x + y, map(lambda x: 1/x, number_digits))

total = reduce(product, [a, b])

n = len(number_digits)
check = total >= n**2

print(f"The results are {total}")
print(f"Is the left handside {a} times {b} greater than or equal to the right␣
 ↪handside?")
print(f"The right hand side which is {n**2} is calculated to show if LHS = RHS,␣
 ↪so therefore, it is {check}.")
```

```
Enter a number 12
The results are 4.5
Is the left handside 3 times 1.5 greater than or equal to the right handside?
The right hand side which is 4 is calculated to show if LHS = RHS, so therefore,
it is True.
```

0.3 Question 6.4

Consider the following functions of two variables

$$x(u,v) = \sin(v)\cos(u)$$
$$y(u,v) = \sin(v)\sin(u)$$
$$z(u,v) = \cos(v)$$

Generate the surface (x, y, z) when $0 \leq u \leq \frac{3\pi}{2}$ and $0 \leq v \leq \pi$.

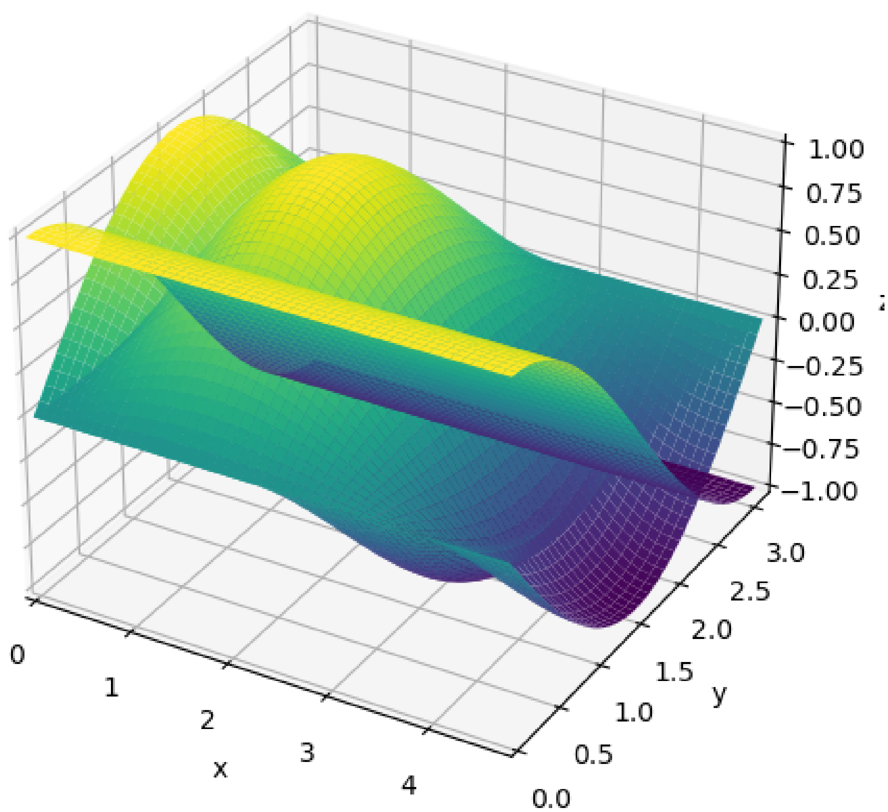
Now consider $x_1(u,v) = \frac{-3}{8}\cos(v)\sin(\frac{4u}{3})$, $y_1(u,v) = \frac{3}{8}\cos(\frac{4u}{3})\cos(v)$ and $z_1(u,v) = -\frac{\sin(v)}{2}$. Generate the surface

$$(\frac{dx_1}{dvdu}, \frac{dy_1}{dudv}, \frac{dz_1}{dv})$$

when $0 \leq u \leq \frac{3\pi}{2}$ and $0 \leq v \leq \pi$. Finally, superimpose these two images.

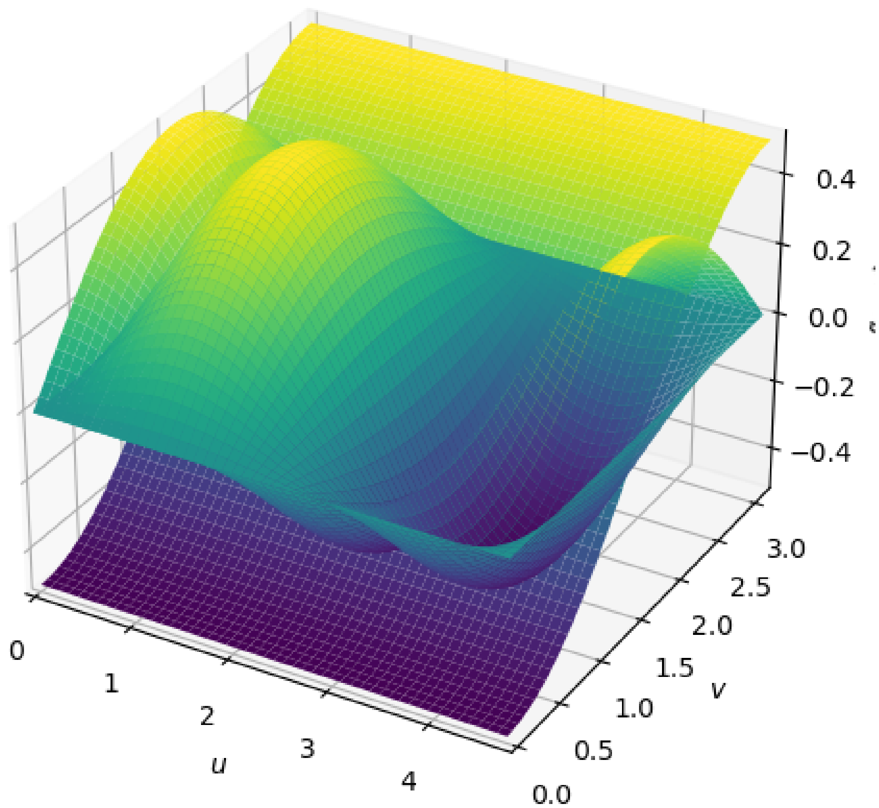
The first part of the question states the $x(u,v), y(u,v) and z(u,v)$ with its respective functions. Next we have to plot a 3D graph with these 2 domains $0 \leq u \leq \frac{3\pi}{2}$ and $0 \leq v \leq \pi$.

[10]:
```python
import sympy as sp

u, v = sp.symbols('u v')

x = sp.sin(v) * sp.cos(u)
y = sp.sin(v) * sp.sin(u)
z = sp.cos(v)

from sympy.plotting import plot3d

surface_plot = plot3d((x, (u, 0, 3*sp.pi/2), (v, 0, sp.pi)), (y, (u, 0, 3*sp.pi/
 ↪2), (v, 0, sp.pi)), (z, (u, 0, 3*sp.pi/2), (v, 0, sp.pi)), show=False)

surface_plot.xlabel = 'x'
surface_plot.ylabel = 'y'
surface_plot.zlabel = 'z'

surface_plot.show()
```



The next part is to superimpose the the plot by doing differentiation in the new $x1$, $y1$ and $z1$ functions and generating the new surface plot of the new functions by differentiating it.

[11]:
```
import sympy as sp

u, v = sp.symbols('u v')

x1 = (-3/8) * sp.cos(v) * sp.sin((4*u)/3)
y1 = (3/8) * sp.cos((4*u)/3) * sp.cos(v)
z1 = -(sp.sin(v)/2)

dx1vu = sp.diff(x1,v,u)
dy1uv = sp.diff(y1,u,v)
dz1v = sp.diff(z1,v)

from sympy.plotting import plot3d

surface_plot2 = plot3d((dx1vu, (u, 0, 3*sp.pi/2), (v, 0, sp.pi)), (dy1uv, (u,
  ↪0, 3*sp.pi/2), (v, 0, sp.pi)), (dz1v, (u, 0, 3*sp.pi/2), (v, 0, sp.pi)),
  ↪show=False)

surface_plot.xlabel2 = 'x'
surface_plot.ylabel2 = 'y'
surface_plot.zlabel2 = 'z'

surface_plot2.show()
```



0.4 Question 7.5

Consider a 10×10 matrix with positve integer numbers as its entries. Write a function θ which is the sum of the third largest number of each row.

Now create a 10×10 matrix M with entry numbers are 1 to 100:

$$M = \begin{pmatrix} 1 & 2 & \dots & 10 \\ 11 & 12 & \dots & 20 \\ \vdots & \vdots & \vdots & \vdots \\ 91 & 92 & \dots & 100 \end{pmatrix}$$

Show $\theta(M)$ is greater than the sum of the numbers in some row.

The first part of the question was to create a 10 by 10 matrix using the `numpy` library, having the numbers from 1 to 100 and using the `reshape` function gives a 10 by 10 matrix.

[12]:
```
import numpy as np

M = np.arange(1, 101).reshape(10, 10)
```

```
print(f"The 10 x 10 matrix from 1 to 100 is")
print(M)
```

```
The 10 x 10 matrix from 1 to 100 is
[[  1   2   3   4   5   6   7   8   9  10]
 [ 11  12  13  14  15  16  17  18  19  20]
 [ 21  22  23  24  25  26  27  28  29  30]
 [ 31  32  33  34  35  36  37  38  39  40]
 [ 41  42  43  44  45  46  47  48  49  50]
 [ 51  52  53  54  55  56  57  58  59  60]
 [ 61  62  63  64  65  66  67  68  69  70]
 [ 71  72  73  74  75  76  77  78  79  80]
 [ 81  82  83  84  85  86  87  88  89  90]
 [ 91  92  93  94  95  96  97  98  99 100]]
```

[13]:
```
def theta(matrix):
    row_sums = []
    for row in matrix:
        row_sorted = sorted(row, reverse=True)
        third_largest = row_sorted[2]
        row_sums.append(third_largest)
    return sum(row_sums)
result = theta(M)
print("Sum of the third largest numbers in each row theta(M):", result)
```

```
Sum of the third largest numbers in each row theta(M): 530
```

I have a written a function called theta where I can find the third largest number in each row and sum them to get the total number. The third largest number in each row are 8, 18, 28, 38, 48, 58, 68, 78, 88 and 98 which adds up to 530.

[14]:
```
sum_first_row = np.sum(M[0])
print("Sum of numbers in the first row:", sum_first_row)

is_greater = result > sum_first_row
print("\nIs theta(M) greater than the sum of numbers in the first row?", 
 ↪is_greater)
```

```
Sum of numbers in the first row: 55

Is theta(M) greater than the sum of numbers in the first row? True
```

To show that $\theta(M)$ is greater than the sum of the numbers in some row, all I had to do was to add all the numbers in the first row from 1 to 10 and the total is 55.

0.5 Question 8.2

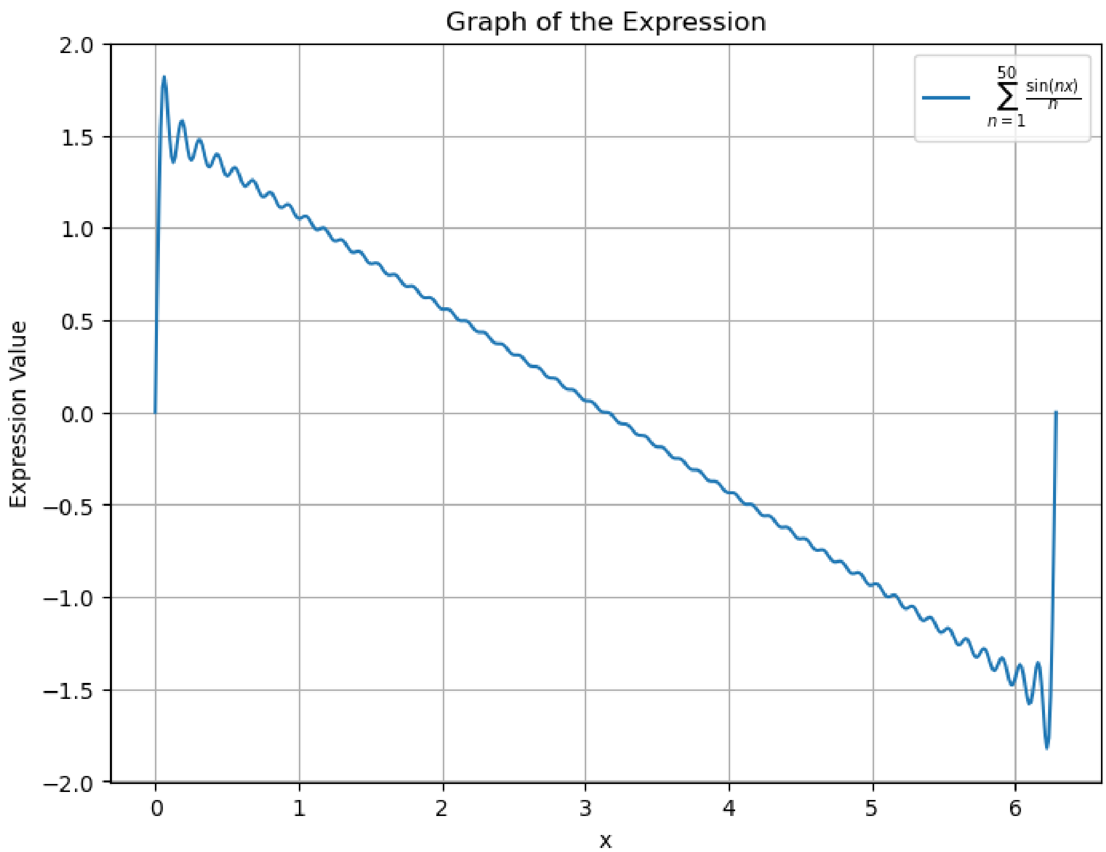
Plot a graph of the expression

$$\sum_{n=1}^{50} \frac{\sin nx}{n}$$

for $0 \leq x \leq 2\pi$.

Adding the functions $\sum_{n=1}^{50} \frac{\sin nx}{n}$ for $0 \leq x \leq 2\pi$ will give.

[15]:
```
import numpy as np
import matplotlib.pyplot as plt

def expression(x, n_max = 50) :
    result = 0
    for n in range(1, n_max + 1) :
        result += np.sin(n*x)/n
    return result

x_values = np.linspace(0, 2 * np.pi, 500)
y_values = expression(x_values)

plt.figure(figsize=(8, 6))
plt.plot(x_values, y_values, label=r'$\sum_{n=1}^{50} \frac{\sin(nx)}{n}$')
plt.title('Graph of the Expression')
plt.xlabel('x')
plt.ylabel('Expression Value')
plt.legend()
plt.grid(True)
plt.show()
```



The graph shows a wiggly line which resembles the $\sin nx$ where $n = 1$. The two straight lines from the beginning and the end shows that there is a hyperbolic function, although no curves, they look identical to each other and comes from the n in the denominator. The number 50 in the function shows that there are 50 waves given in the sum function. And the sum function gives us an almost straight line with a mix of sine function and hyperbolic function.